Prateek Tripathi

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Summary

Software Developer with strong expertise in scalable web architectures, REST API development, and SQL-driven backend systems. Experienced in designing microservices-based applications, cloud deployments on AWS, and implementing secure coding standards. Familiar with ASP.NET MVC architecture, C, and software development lifecycle (SDLC) methodologies. Proficient in system design, database query optimization, and building distributed, high-performance applications.

Education

Sinhgad Academy of Engineering, Pune *June 2020 – June 2024*
B.E. in Computer Engineering

GPA: 8.36 / 10.0
Coursework: Computer Architecture, Data Structures and Algorithms (DSA), Machine Learning & AI, Operating Systems, Database Management Systems (DBMS), Theory of Computation, Software Engineering, Computer Networks

Experience

Software Team Lead *Bangalore, India*
Satyukt Analytics *April 2026 – Present*

- Promoted to Software Team Lead to drive technical strategy, system architecture, and team execution for agri-tech platforms.
- Leading a team of developers to build scalable, data-driven systems powered by satellite and geospatial datasets.
- Architecting backend systems using Python (FastAPI/Flask), Node.js, and microservices-based design.
- Designing and optimizing cloud infrastructure on AWS (EC2, Lambda, S3, RDS) focusing on scalability, reliability, and cost efficiency.
- Managing containerized deployments using Docker and Kubernetes (EKS/ECS) for distributed and cloud-native systems.
- Mentoring developers, conducting code reviews, and establishing engineering best practices for maintainable and high-quality code.
- Collaborating with data science and remote sensing teams to deliver high-performance analytics and data transformation pipelines.
- Improving system performance, monitoring, and observability using CloudWatch and logging systems.

Software Engineer *Bangalore, India*
Satyukt Analytics *Dec 2025 – March 2026*

- Designed and maintained scalable full-stack and data-driven systems for satellite-based agri-tech platforms.
- Developed backend APIs using Python (Flask/FastAPI) and built data ingestion pipelines for large-scale geospatial datasets.
- Implemented ETL workflows for structured and semi-structured data, improving data processing efficiency.
- Deployed and optimized cloud workloads on AWS (EC2, Lambda, S3, RDS) ensuring high availability and performance.
- Containerized microservices using Docker and Kubernetes for scalable deployments.
- Explored distributed data processing using Apache Spark / Databricks for large-scale analytics.

Research Assistant (JRF) *Raipur , India*
National Institute of Technology (NIT) Raipur *Aug 2024 – Nov 2025*

- Led a team of developers to build an automation platform for finance, improving GST filing workflows by 60
- Conducted code reviews and technical guidance for junior contributors.
- Integrated REST APIs with React/Next.js frontends, ensuring optimized rendering and async handling.
- Engineered a chatbot-driven system using Express.js, Next.js, Kafka, Docker, Kubernetes, and AWS (EC2, S3) to support intelligent financial operations.
- Architected a secure, scalable, and cloud-native infrastructure ensuring high availability and compliance with financial standards.

Projects

FINAGER – Fintech Automation Suite *GitHub Repository*

- Built a scalable GST filing platform with chatbot-driven UI, reducing manual filing time by 60%.
- Developed secure, cloud-native infrastructure using Next.js, Express.js, Kafka, Docker, and AWS (EC2, S3).
- Implemented modular microservices to handle tax logic, chatbot NLP, and file uploads with high concurrency.

KPI Dashboard – Raipur District Administration *Live Website*

- Designed a KPI monitoring platform for 300+ officers with interactive scoring and geo-tagged photo logging.
- Developed REST APIs and reusable components using Next.js, React Native, and MongoDB.
- Stress on React UI optimization, state management (Zustand/Context API), real-time data visualization.

Swachhta App – Nagar Nigam Raipur *GitHub Repository*

- Developed a mobile app using YOLO for real-time detection of garbage sites and auto-reporting.
- Built with React Native, Express.js, Python, and AWS (S3, Lambda); integrated photo upload + scoring.
- Enabled Nagar Nigam to track cleanliness reports from citizens across 6+ wards.

Skills

Languages: Java, Python, SQL, TypeScript, C++, C, C# (Working Knowledge), DSA

Technologies:
Web & .NET: ASP.NET MVC (Conceptual Knowledge), .NET Fundamentals, LINQ (Exposure)
Data Engineering: Apache Spark (PySpark), Databricks (Hands-on / Learning), ETL Pipeline Design, Data Processing, Batch & Streaming Fundamentals, Data Lake Concepts
Frontend: HTML, CSS, JavaScript, React, Next.js
Backend & System Design: Node.js, Express.js, Python (FastAPI, Flask), REST APIs, Microservices Architecture, Design Patterns (MVC, Repository – Conceptual), Secure Coding Practices, SDLC (Agile)
Database & Data Modelling: SQL (Joins, Indexing, Query Optimization), MongoDB, Schema Design, Stored Procedures (Basic Knowledge)
DevOps: Docker, Kubernetes, GitHub Actions, CI/CD Pipelines
Cloud: AWS (EC2, S3, Lambda, RDS, CloudWatch), Firebase
AI/ML: BERT, YOLO, Image Segmentation, Applied Machine Learning
Streaming & Messaging: Apache Kafka, Event Driven Architecture
Other Tools: JWT, Postman, Git, Nginx

Certificates

- **Java Certification** — Completed in 08/2022 – 09/2022
- **C++ Certification** — Completed in 04/2022 – 05/2022
- **C Certification** — Completed in 02/2022 – 03/2022